Linear Independent

Remark: If $S=\{\underline{s}_1, \cdots, \underline{s}_k\}$ is finite,

then S is linearly indep.

~~(~~$\Leftrightarrow$ all $\underline{S}' \subseteq S$ is linearly indep.~~)~~

$\Leftrightarrow$ the only sol.

of $\alpha_1 \underline{s}_1 + \cdots + \alpha_k \underline{s}_k = 0$

is $\alpha_1 = \cdots = \alpha_k = 0$

E.g., $V = C^\infty(\mathbb{R})$, ① $S = \{e^x, \cos(x)\}$

To see if S is l.i.,

consider $\alpha e^x + \beta \cos(x) = 0$ (∵ 取 $x=0$ 和 $x=1$ ①

$\downarrow$

$(\alpha e^x + \beta \cos(x) = \underline{0}(x) = 0, \ \forall x \in \mathbb{R})$

② $S' = \{\cos^2(x), \sin^2(x), 1\}$ ∵ $1(x) = 1, \ \forall x$

S' is l.d. $\Leftarrow$ ① $\cos^2(x) +$ ① $\sin^2(x) -$ ① $\cdot 1(x) = 0$

Basis & Dimensions

Basis

Def.: a subset S is a basis of V if ① S spans V & ② S is l.i.

E.g., ① $\{e_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow i\text{th coord.} \mid i = 1, \cdots, n\}$

$\downarrow$

$V = \mathbb{R}^n$

② all polynomials

$\downarrow$ can be written as

a (finite) l. combination of $S = \{1, x, x^2, \cdots\}$

③

④ $V = C^\infty(\mathbb{R})$, $S = \{1, x, x^2, \cdots\}$

S is l.i., but $Span(S) \neq V$

e.g., $\cos x \in V$

Taylor expansion $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \notin Span(S)$

∞ terms

⑤ There can be different bases for a v. space V

e.g. If $A = \{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$,

then $B = \{\underline{v}_1 - \underline{v}_2, \underline{v}_2 - \underline{v}_3, \underline{v}_3\}$,

Proof:

(a) 先正 $Span(B) = V$

$\underline{v}_3 \in Span(B)$

$\underline{v}_2 = (\underline{v}_2 - \underline{v}_3) + \underline{v}_3 \in Span(B)$

$\underline{v}_1 = \cdots \in Span(B)$

$\Rightarrow Span(A) \subseteq Span(B)$

(Prop. 1.1 ---⑥) $\Rightarrow V \subseteq Span(A)$ ---⑦

(b) B is l.i.

Consider $\alpha(\underline{v}_1 - \underline{v}_2) + \beta(\underline{v}_2 - \underline{v}_3) + \gamma\underline{v}_3 = 0$

$\alpha\underline{v}_1 + (-\alpha+\beta)\underline{v}_2 + (-\beta+\gamma)\underline{v}_3 = 0$

$$\begin{cases} \alpha = 0 \\ \beta - \alpha = 0 \\ \gamma - \beta = 0 \end{cases} \Rightarrow$$

Proposition 1.4 (1.9 on the slide)

If S is a basis of V,

then $\forall \underline{v} \in V$ can be expressed uniquely

as $\underline{v} = \alpha_1 \underline{v_1} + \cdots + \alpha_n \underline{v_n} \in \mathrm{Span}(S)$

Meaning of "uniquely":

If $\underline{v} = \alpha_1 \underline{s_1} + \cdots + \alpha_p \underline{s_p}$ $(\forall i, \underline{s_i} \in S)$

and $\underline{v} = \beta_1 \underline{s'_1} + \cdots + \beta_q \underline{s'_q}$ $(\forall i, \underline{s'_i} \in S)$

in $\mathrm{Span}(S)$,

then $p = q$

& $\forall i, \underline{s_i} = \underline{s'_i}$

& $\forall i, \alpha_i = \beta_i$

Proof:

Suppose that

— holds for 2 finite subsets

$B = \{\underline{s_1}, \cdots, \underline{s_p}\}$ and $B' = \{\underline{s'_1}, \cdots, \underline{s'_q}\}$

then we have

common

$B = \{\underline{s_1}, \cdots, \underline{s_v}, \underline{s_{v+1}}, \cdots, \underline{s_p}\}$

no common terms

$B' = \{\underline{s'_1}, \cdots, \underline{s'_v}, \underline{s'_{v+1}}, \cdots\cdots, \underline{s'_q}\}$

(见下页)

证明 unique expression

Then $\sim$ implies that

(★) ($R_{\sim}$)

Since S is a basis

(Remark: the proof of $\Rightarrow$ when $|S|=\infty$ is harder)

Finitely Generated

Def.: A v. space V is finitely generated if $\exists$ a finite subset $S = \{\underline{s_1}, \cdots, \underline{s_k}\}$, s.t. $V = Span(S)$ ← (S is not necessarily be l.i.)

e.g. $\mathbb{R}^n = Span\{\underline{e_1}, \cdots, \underline{e_n}\}$ is f.g.

$M_{m\times n}(\mathbb{R}) = Span\{E_{ij} \mid \begin{matrix} 1\le i\le m \\ 1\le j\le n \end{matrix}\}$ is f.g.

(E_{ij}: (i,j) entry 為 1)

☆ However $V = P(\mathbb{R})$ is not f.g.

Suppose on contrary

$V = Span\{$

Let $M = max(n_1, n_2, \cdots, n_k)$

Then x^{M+1}

Theorem Let V be f.g., then V must have a basis.

Proof: ① If $\{\underline{v_1}, \cdots, \underline{v_k}\}$ is l.i., then it is a basis of V

② If $\cdots$ is l.d.

then $\alpha_1 \underline{v_1} + \cdots + \alpha_k \underline{v_k} = 0$, for $\alpha_1, \cdots, \alpha_k$ are not all zeros

W.L.O.G. (by rearranging the terms) assume $\alpha_1 \neq 0$,

then $\underline{v_1} = \left(-\frac{\alpha_2}{\alpha_1}\right)\underline{v_2} + \cdots + \left(-\frac{\alpha_k}{\alpha_1}\right)\underline{v_k}$

$\in \text{Span}\{\underline{v_2}, \cdots, \underline{v_k}\}$

$\therefore \{\underline{v_1}, \underline{v_2}, \cdots, \underline{v_k}\} \in \text{Span}\{\underline{v_2}, \cdots, \underline{v_k}\}$

$\Rightarrow \text{Span}\{\underline{v_1}, \underline{v_2}, \cdots, \underline{v_k}\} \subseteq \text{Span}\left(\text{Span}\{\underline{v_2}, \cdots, \underline{v_k}\}\right)$

This will

Remark: Any v. space V has a basis.

Unlike the f.g. basis, ---